



DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Sem)

(CIE I- B. E. PROGRAMS, EC, EE, EI, ET)

Date	21 October 2024	Maximum Marks	10 +50
Time	09:30 AM to 11 :30 AM	Duration	120 Mins
Semester	III Sem B. E. (EC, EE, EI, ET)		
MA231TA - LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS			

Q. No.	PART - A	M	BT	CO														
1	The first four moments of distribution about the value 5 of the variable are 0, 24, 32 and 45. Calculate the mean and variance.	2	L2	2														
2	In a partially destroyed laboratory record of an analysis of a correlation data, the following results were noted: variance of $x = 9$, equations of lines of regression of y on x is $4x - 5y + 33 = 0$ and x on y is $20x - 9y = 107$. For the given data the value of standard deviation of $y =$ _____ and $\tan\theta =$ _____.	2	L3	1														
3	Find $(\bar{x}, \bar{y})$, when the equations of the regression lines are $2x + 3y + 1 = 0$ and $x + 6y = 4$.	2	L1	2														
4	The following data shows ranks of 6 students in two subjects, then rank correlation coefficient is _____. <table border="1"><tr><td>Mathematics</td><td>5</td><td>1</td><td>3</td><td>6</td><td>2</td><td>4</td></tr><tr><td>Physics</td><td>6</td><td>1</td><td>5</td><td>4</td><td>3</td><td>2</td></tr></table>	Mathematics	5	1	3	6	2	4	Physics	6	1	5	4	3	2	2	L1	2
Mathematics	5	1	3	6	2	4												
Physics	6	1	5	4	3	2												
5	The corresponding normal equations of the multi linear regression model $y = a + bx_1 + cx_2 + dx_3$ are _____.	2	L1	1														

PART – B											
1	Calculate the first four moments about mean, coefficient of skewness and coefficient of kurtosis for the following frequency distribution.							10	L3	2	
	x	5	10	12	14	15	19				22
	f	1	2	3	12	4	5				3
2a	The standard deviation of distribution is 4. What must be the value of the fourth moment about the mean in order that the distribution be (i) Leptokurtic (ii) Mesokurtic and (iii) Platykurtic.							4	L3	1	



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2b	Seven students got the following percentage of marks in two subjects x and y . Compute their rank correlation coefficient.	6	L2	2																					
	<table><tr><td>Marks in x</td><td>68</td><td>64</td><td>75</td><td>64</td><td>75</td><td>40</td><td>64</td></tr><tr><td>Marks in y</td><td>62</td><td>58</td><td>68</td><td>81</td><td>68</td><td>48</td><td>70</td></tr></table>	Marks in x	68	64	75	64	75	40	64	Marks in y	62	58	68	81	68	48	70								
Marks in x	68	64	75	64	75	40	64																		
Marks in y	62	58	68	81	68	48	70																		
3	The following table gives the stopping distance y in meters of a motor bike moving at a speed of x Kms/hour when the breaks are applied. <table><tr><td>x</td><td>16</td><td>24</td><td>32</td><td>40</td><td>48</td><td>56</td></tr><tr><td>y</td><td>0.4</td><td>0.7</td><td>1.3</td><td>1.9</td><td>2.7</td><td>3.8</td></tr></table> Find the coefficient of correlation between the speed and stopping distance and estimate the regression line of stopping distance on speed and speed on stopping distance.	x	16	24	32	40	48	56	y	0.4	0.7	1.3	1.9	2.7	3.8	10	L3	3							
x	16	24	32	40	48	56																			
y	0.4	0.7	1.3	1.9	2.7	3.8																			
4	Determine the equation of the regression plane connecting x_1, x_2, and y given <table><tr><td>Diffusion time(hours) x_1</td><td>1.5</td><td>2.5</td><td>0.5</td><td>1.2</td><td>2.6</td><td>0.3</td></tr><tr><td>Sheet resistance (Ohms-cm) x_2</td><td>66</td><td>87</td><td>69</td><td>141</td><td>93</td><td>105</td></tr><tr><td>Current gain (y)</td><td>5</td><td>8</td><td>7</td><td>9</td><td>10</td><td>9</td></tr></table> Estimate y at $x_1 = 1.8$ and $x_2 = 112$.	Diffusion time(hours) x_1	1.5	2.5	0.5	1.2	2.6	0.3	Sheet resistance (Ohms-cm) x_2	66	87	69	141	93	105	Current gain (y)	5	8	7	9	10	9	10	L3	3
Diffusion time(hours) x_1	1.5	2.5	0.5	1.2	2.6	0.3																			
Sheet resistance (Ohms-cm) x_2	66	87	69	141	93	105																			
Current gain (y)	5	8	7	9	10	9																			
5	Let $\mathbb{R}^2, \mathbb{R}^3, M_{2 \times 2}$ (set of all 2×2 real matrices) be vector spaces with usual addition and scalar multiplication. Verify whether the following sets forms a subspace or not. Justify your answer. (i) $s_1 = \{(x, y, z) \in \mathbb{R}^3 : x - 3y + 4z = 0\}$ (ii) $s_2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ (iii) $s_3 = \{A \in M_{2 \times 2} : \det(A) = 0\}$ (iv) $s_4 = \{(x, y) \in \mathbb{R}^2 : \text{set of all lines passing through the origin}\}.$	10	L2	2																					

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	4	6	-	-	6	2	2	-	-	-
	Test		4	26	20	-	-	16	34	-	-	-
